

Report 4: Neural Network

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1 Network Architecture

The neural network was implemented from scratch using object oriented programming.

The implementation contains 3 classes:

- Neuron
- Layer
- NeuralNetwork

The network architecture is read from a text file.

Example:

3
2
2
1

Meaning:

- 3 layers
- 2 input neurons
- 2 hidden neurons
- 1 output neuron

Each neuron contains:

- weights
- bias
- sigmoid activation function

2 Feedforward

Each neuron computes a linear sum:

$$z = \sum x_i w_i + b$$

Then applies the sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The outputs of one layer become the inputs of the next layer.

Feedforward is performed layer by layer until the final output is obtained.

3 XOR Experiment

The XOR neural network was initialized using the weights and biases from the lecture slides.

Hidden layer:

$$[-10, -10], \quad b = 15$$

$$[10, 10], \quad b = -5$$

Output layer:

$$[10, 10], \quad b = -15$$

Inputs tested:

- (0,0)
- (0,1)
- (1,0)
- (1,1)

Results:

[0,0] -> 0.007
[0,1] -> 0.992
[1,0] -> 0.992
[1,1] -> 0.007

The neural network successfully approximates the XOR function.

```
namle@192:~/dl2026/Lab4$ source dl2026/bin/activate
bash: dl2026/bin/activate: No such file or directory
namle@192:~/dl2026/Lab4$ cd ..
namle@192:~/dl2026$ source dl2026/bin/activate
(dl2026) namle@192:~/dl2026$ cd Lab4
(dl2026) namle@192:~/dl2026/Lab4$ python neural_network.py
[0, 0] -> 0.42436377623451826
[0, 1] -> 0.4365732064532725
[1, 0] -> 0.4365732064532725
[1, 1] -> 0.42436377623451826
(dl2026) namle@192:~/dl2026/Lab4$ python neural_network.py
[0, 0] -> 0.007152788188896492
[0, 1] -> 0.9923558641717396
[1, 0] -> 0.9923558641717396
[1, 1] -> 0.007152788188896487
(dl2026) namle@192:~/dl2026/Lab4$
```

Figure 1: Results of the XOR neural network